ERA 5 calibration to Elexon power

S. Gomez[1]

[1]School of Mathematics, University of Edinburgh

2026-01-13

Table of contents I

1 Overview

2 Calibration of generic power curve estimate

3 Power curve modelling

4 Appendix

Overview

Updates

New

- Power curve modelling
- Calibration comparison
- Spatial correlation check

Next steps

- Wind speed Cut-off estimation (pooling data)
- Wind direction effects on power curve
- Out-of-sample validation
- Compare model against benchmarks: Quantile mapping, GAMs
- Calibration spatiotemporal model

Data sources

Wind speed

- ERA5 at wind farms
  - Hourly data
  - Spatial resolution 0.25° x 0.25°
  - 10m and 100m heights

Wind power

- Elexon BMU data (since 2019)
  - Half hourly data
  - Generation, curtailment, potential, capacity
  - Outage data (REMIT)
- REPD database
  - Location, turbine height, capacity

Overview

1. ERA 5 to wind farm

Vertical interpolation to turbine height h.

$$w(h) = w_{100}\left(\frac{h}{100}\right)^{\alpha}, \text{ where } \alpha = 1/7$$

2. Wind speed to power

Generic power curves rescaled to wind farm capacity.

$$\hat{PC}_i(w) = PC_i(w) \times \frac{C_i}{\text{Rated power}},$$

where C_i is the capacity at location i

Overview of power conversion

3. Potential generation

Curtailment and outages are two main events that impact observed generation o_{it}

- Curtailment is added giving rise to potential generation:

$$p_{it} = o_{it} + \text{curt}_{it}$$

- Outage data shows additional limits on capacity
- Currently outage periods are excluded

Calibration

4. Calibration of generic power curve estimate

ERA5-derived power estimate $\hat{p}_{it}$ is compared versus potential power p_{it}

$$\hat{p}_{it} = \hat{PC}_i(w_{it})$$
$$p_{it} = \beta_0 + \beta\hat{p}_{it} + s_i + u_t,$$

where s_i and u_t represent spatial and temporal effects.

Calibration of generic power curve estimate

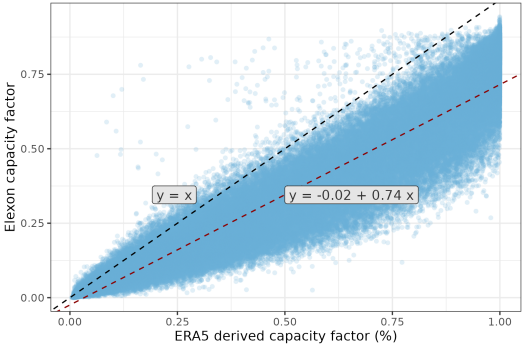
GB level aggregation

- Power aggregated at GB level

$$p_{\text{tot},t} = \sum_i p_{it} / \sum_i C_i$$

- Each point represents one hour
- Overestimation persits but dispersion is lower now

Scatter aggregated power

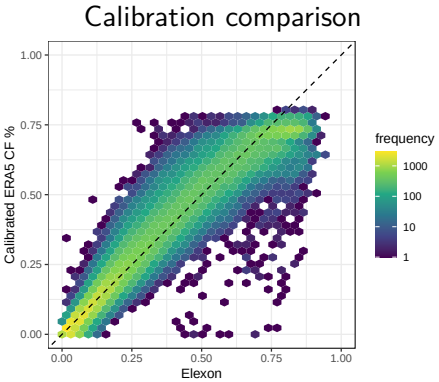


Initial linear model

- Features of importance:
  - m: month
  - k: Type (offshore / onshore)

$$p_t^{calib} = \alpha_{k,m} + \beta_{k,m}\hat{p}_t,$$

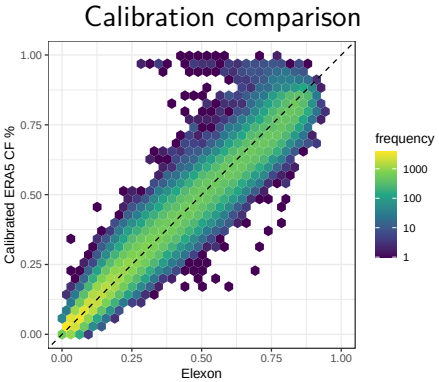
where $\alpha_{k,m}$ and $\beta_{k,m}$ represent the intercept and slope, varying by type and month



Updated linear model

- Using step AIC criteria selection:
  - k: Type (offshore / onshore)
  - m: month, h: hour
  - w_t: wind speed

$$p_t^{calib} = \alpha_{k,m,h} + \beta_{k,m}\hat{p}_t + \sum_{j=1}^{3} \phi_j w_t^3$$



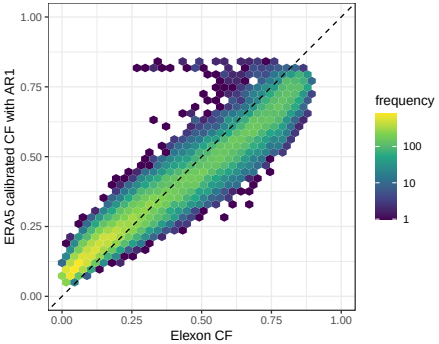
Autoregressive error

- Using step AIC criteria selection:
  - k: Type (offshore / onshore)
  - m: month, h: hour
  - w_t: wind speed
  - u_t: AR1

$$p_t^{calib} = \alpha_0 + \sum \alpha_e + \beta_k \hat{p}_t + f_w(w_t) + u_t$$

where α_0 is the intercept, the α_e $e \in k, m, h$ represent random effects (cyclic for month and hour), β_k is a random slope, f_w is smooth effect (RW2).

Calibration comparison



Power curve modelling

Power curve modelling

Components

- Power curve estimate from data
- Probability of zero generation
- Penalisation to make power curve resemble manufaturer's PC

Using a three-likelihood approach for power curve estimation:

- **1st likelihood:** Observed power is modelled as a smooth function of wind speed
- **2nd likelihood:** Zero inflated component
- **3rd likelihood** Penalisation towards a generic PC

Power curve model equations

Model is a Zero Inflated Beta (ZIB) with penalisation component:

Zero–probability model (Bernoulli)

$$\eta_i^{(0)} = \alpha_{\mathrm{bern}} + f_{\mathrm{bern}}(w_i;\, \mathbf{s}(i))\,, \tag{1}$$

$$Z_i \mid \eta_i^{(0)} \sim \mathrm{Bernoulli}(p_i), \qquad p_i = \mathrm{logit}^{-1}\left(\eta_i^{(0)}\right), \tag{2}$$

Positive–output model (Beta)

$$\eta_i^{(\beta)} = \alpha_{\mathrm{pc}} + f_{\mathrm{pc}}(w_i;\, \mathbf{s}(i))\,, \tag{3}$$

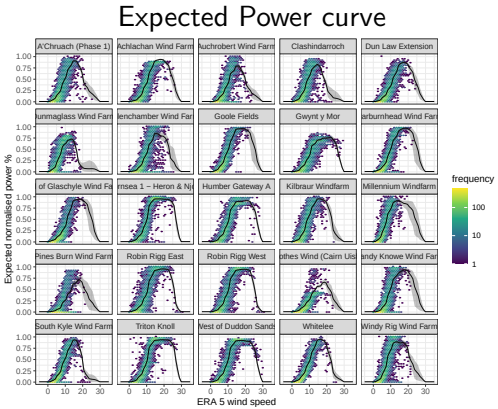
$$P_i^{\mathsf{obs}} \mid \eta_i^{(\beta)} \sim \mathrm{Beta}(\mu_i\,, \phi), \qquad \mu_i = \mathrm{logit}^{-1}\left(\eta_i^{(\beta)}\right), \tag{4}$$

Pseudo–likelihood model (Gaussian)

$$\tilde{P}_i \mid \eta_i^{(\beta)} \sim \mathcal{N}\left(\eta_i^{(\beta)}, \tau_{\mathrm{ps}}^{-1}\right), \qquad \tau_{\mathrm{ps}} = \texttt{precision or penalisation.} \tag{5}$$

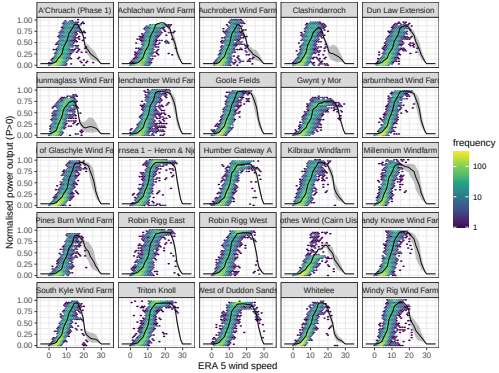
Power curve model estimates

- Starting models with
  - 1Y of data (2024)
  - 25 wind farms
  - Hourly data
  - Excluding outages listed in REMIT

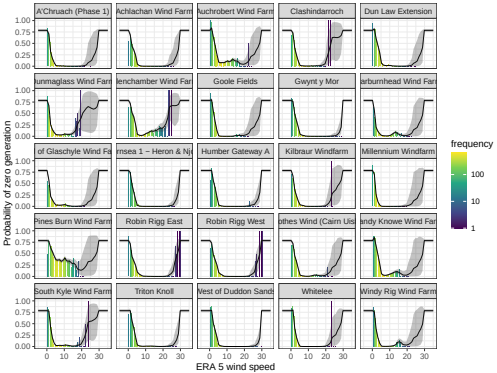


Power curve model estimates

Power curve estimate



Probability of zero generation

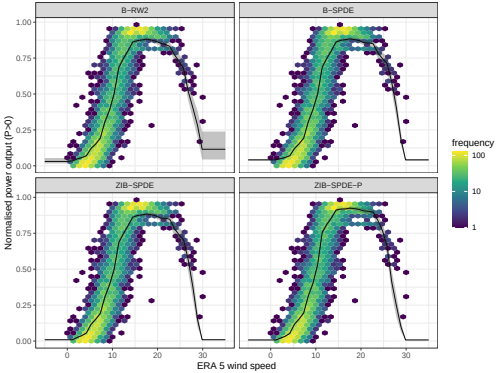


Alternative versions

- Beta model with RW2 power curve (B-RW2)
- Beta model with 1D SPDE power curve (B-SPDE)
- ZIB model with 1D SPDE power curve (ZIB-SPDE)
- ZIB model with 1D SPDE power curve penalised (ZIB-SPDE-P)

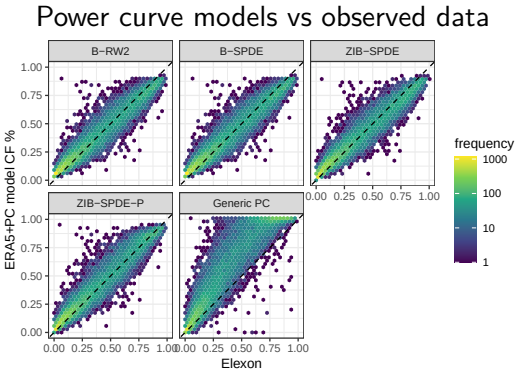
Zero Inflated models better capture low wind power regions.

Power curve estimates under different models



ZIB model vs observed data

- Power curve modelling highly improves the calibration
- ZIB models deal better with the tail regions
- Penalised version actually pulls estimate toward Generic PC
- We could pull estimate instead to a 5 parameter curve estimate



Error metrics

Error metrics calculated on 30% of the sampled wind farms data.

model	RMSE	MAE	Bias
Generic PC	0.2188	0.1583	0.1469
B-RW2	0.0995	0.0723	0.0191
B-SPDE	0.0994	0.0722	0.0190
ZIB-SPDE	0.0976	0.0691	0.0103
ZIB-SPDE-P	0.1016	0.0694	0.0269

Next Steps

- Combine updated power curve model with calibration
- Wind speed Cut-off estimation (pooling data)
- Wind direction effects on power curve
- Out-of-sample validation
- Compare model against benchmarks: Quantile mapping, GAMs
- Calibration spatiotemporal model

Appendix

Previous research

Power curve modelling

Binning method

$$P_i = \frac{1}{n_i} \sum_{j=1}^{n_i} P_{ij}$$

where: P_{ij} is the j th power observation in bin i and n_i no. of observations in bin i

Logistic

$$P(u) = a \frac{1 + m \exp(-u/\tau)}{1 + n \exp(-u/\tau)}$$

where a represents the upper asymptote, n, m shape the lower asymptote, and τ controls the transition.

Power curve modelling

5 parameter curve

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

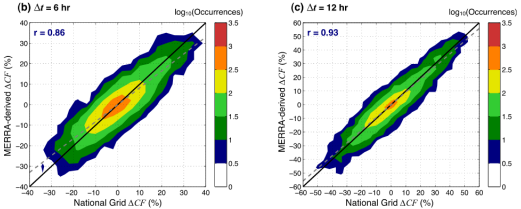
where: A and D are the upper and lower asymptotes, C is the inflection point, B the slope at inflection point, adn G controls the asymmetry.

Reanalysis data to quantify extreme wind power statistics

D. Cannon, D. Brayshaw, et. al (2015)

- MERRA wind speed validated with MIDAS
- Vertical interpolation with a logarithmic change
- Calibrated power curves based on manufacturers PC
- Use that to analys extreme low and high levels, and ramps

ΔCF comparison

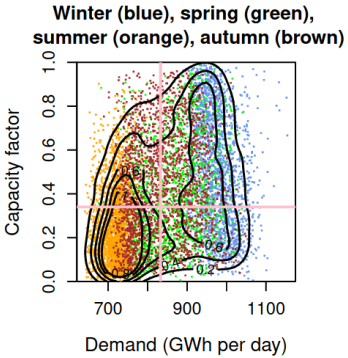


Balancing energy

H. Thornton, D. Brayshaw (2020) analyse the relationship between weather, energy demand, and wind power.

- ERA Interim wind speed
- Cubic power curve with air density correction
- CF compared with GB average from other studies
- Seasonal effects on Demand and

ΔCF comparison

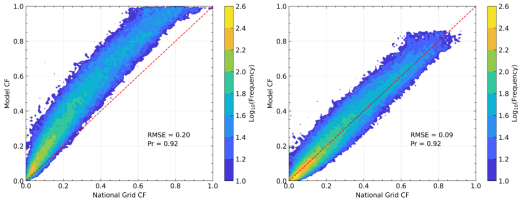


Analysis of extreme wind droughts

Panit Potisomporn, C. Vogel (2024) perform a extreme value analysis of wind droughts in GB.

- Use ERA5 wind speeds calibrated to MIDAS with QM
- Build a ML algorithm that learns how to extrapolate wind speed from 10m to hub height
- Use a 5 parameter logistic function to model power curve
- Model energy losses with factors by type.

CF calibration



References

Cannon, D. J., D. J. Brayshaw, J. Methven, P. J. Coker, and D. Lenaghan. 2015. "Using Reanalysis Data to Quantify Extreme Wind Power Generation Statistics: A 33 Year Case Study in Great Britain." *Renewable Energy* 75 (March): 767–78. https://centaur.reading.ac.uk/38448/.

Potisomporn, P. 2024. "Spatiotemporal Variability of Extreme Low Wind Power Events in Great Britain - ORA - Oxford University Research Archive." Http://purl.org/dc/dcmitype/Text. https://ora.ox.ac.uk/objects/uuid:de37bee2-1813-42e5-a69c-9b15f038d096.

Thornton, Hazel E. 2020. "Atmospheric Circulation, Seasonal Predictability and Britain's Energy System." PhD thesis, University of Reading. https://doi.org/10.48683/1926.00095832.